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G. Can You Split the Array?

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given an array containing N positive integers. Your task is to partition the array into at most K subarrays so that the maximum sum among all subarrays is as small as possible.

Input

The first line contains two integers N ($1 \leq N \leq 2 \times 10^5$) and K ($1 \leq K \leq N$), the size of the array and the number of subarrays in the partition.

The next line contains N integers $a_1, a_2, \dots, a_N$ ($1 \leq a_i \leq 10^6$), the elements of the array.

Output

Print one integer: the minimum possible largest subarray sum.

Examples

input	Copy
5 3 1 2 3 4 5	
output	Copy
6	

input	Copy
5 3 1 5 1 1 5	
output	Copy
6	

Note

Brief Explanation for First Sample: There are many ways to partition the array

- The optimal/best partition is $[1, 2, 3], [4], [5]$. The maximum sum for [this partition](#) is 6. This is because
 - the sum of the first subarray $[1, 2, 3]$ is $1 + 2 + 3 = 6$;
 - the sum of the second subarray $[4]$ is 4; and
 - the sum of the third subarray $[5]$ is 5.
- Another partition is $[1, 2], [3, 4], [5]$. The maximum sum for [this partition](#) is 7. This is because
 - the sum of the first subarray $[1, 2]$ is $1 + 2 = 3$;
 - the sum of the second subarray $[3, 4]$ is $3 + 4 = 7$;
 - the sum of the third subarray $[5]$ is 5.

CSE221 Assignment 02 Summer 2026

Contest is running

6 days

Contestant



→ Submit?

 Language: Java 21 64bit

Choose file:

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Success!



Since we want to find the smallest possible value for the maximum sum, the first partitioning ($[1, 2, 3]$, $[4]$, $[5]$) is the best. The answer is 6.

Problem Extension: What if we allow negative numbers? How would the solution change?

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The only programming contests Web 2.0 platform

Server time: Jun/29/2026 16:49:15^{UTC+6} (i1).

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